Synthetic Inverse PINN Experiments for Lynx and Hare Dynamics

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1 Inverse inference with a physics informed neural network

This section tests whether an inverse PINN can recover known parameters and produce ODE forecasts under controlled observation noise. Every observation and result in this section is synthetic. The year labels mirror the planned 1900–1915 training and 1916–1920 testing split; they are not historical pelt counts.

1.1 Model and normalized approximation

We use the Rosenzweig–MacArthur model [3],

$$\dot{H} = rH(1 - H/K) - \frac{aHP}{b + H}, \tag{1}$$

$$\dot{P} = \frac{eaHP}{b + H} - mP. \tag{2}$$

Time is measured in years and both states are synthetic abundance indices. The primary inverse task estimates r, a, m, with $K = 60$, $b = 20$ and $e = 0.6$ fixed. These constants equal their generating values only in the controlled synthetic study; this favorable assumption must be revisited for historical observations.

Let s_H, s_P be the maxima of the training observations and $T = 15$ years. A network with three hidden layers of 48 tanh units maps $x = 2t/T - 1$ to positive normalized states $(u_\theta, v_\theta) = \text{softplus}(N_\theta(x)) + 10^{-10}$, giving $\widehat{H} = s_H u_\theta$ and $\widehat{P} = s_P v_\theta$. Unknown rates are bounded by $q = \ell_q + (h_q - \ell_q)\,\text{sigmoid}(\eta_q)$, using $r \in [0.2, 2.5]$, $a \in [0.3, 4]$ and $m \in [0.1, 1.2]$. These are numerical study bounds, not biological conclusions.

1.2 Physics residuals and optimization

Following the data-plus-physics construction of PINNs [2], we form

$$R_H = \frac{du_\theta}{dt} - \frac{f_H(s_H u_\theta, s_P v_\theta; q)}{s_H}, \tag{3}$$

$$R_P = \frac{dv_\theta}{dt} - \frac{f_P(s_H u_\theta, s_P v_\theta; q)}{s_P}, \tag{4}$$

$$L = L_{\text{data}} + 10L_{\text{physics}} + 10L_{\text{IC}}, \qquad L_{\text{physics}} = \frac{1}{2N_c}\sum_{j=1}^{N_c}(R_H(t_j)^2 + R_P(t_j)^2). \tag{5}$$

Data loss is the mean squared normalized observation error across both species; IC loss is the two-state mean squared error at $t = 0$. The first observation appears in both terms, deliberately emphasizing the initial state. There are $N_c = 161$ equally spaced collocation points on $[0, 15]$,

including endpoints, and none in the forecast interval. PyTorch automatic differentiation [1] differentiates with respect to elapsed time, accounting for the internal factor $2/T$.

Training uses float64, one CPU thread and fixed seeds. Xavier initialization and a 500-step data/IC warmup precede Adam (learning rate 0.002, at most 5000 joint steps, gradient norm clipped at 10). Complete training losses are checked every 100 steps; Adam stops after 15 checks without improvement exceeding 10^{-8} or at its cap. The best state is refined with L-BFGS and strong-Wolfe line search, at most 600 iterations and 750 closure evaluations, with gradient and change tolerances 10^{-10} and 10^{-12}. The saved state minimizes the recorded training objective. Closure logs include line-search trials, so they need not be monotonic. An iteration cap does not prove global convergence. An offset grid of 307 training-domain points diagnoses residuals between collocation points after fitting.

1.3 Synthetic validation and forecasting protocol

The generating parameters are $(r, K, a, b, e, m) = (1, 60, 2, 20, 0.6, 0.5)$, with $(H_0, P_0) = (35, 9)$. Multiplicative noise uses $y_i^{\text{obs}} = y_i^{\text{true}}(1 + 0.03\epsilon_i)$, $\epsilon_i \sim \mathcal{N}(0, 1)$ independently per species and time. Values are clipped below at 10^{-8} and the initial state is exact by design. Noise seed 314 is fixed. We compare 61 clean quarterly, 61 noisy quarterly and 16 noisy annual training samples, each with optimizer seeds 17, 29 and 43. These optimizer replicates condition on one noise realization per case and are not uncertainty intervals.

All ODE integrations use SciPy RK45 [4], relative tolerance 10^{-8}, absolute tolerance 10^{-10} and maximum step 0.1 years. A six-start least-squares benchmark uses the same unknowns, fixed values, bounds, state scaling, initial condition and solver. It minimizes normalized training observation errors; an additional physics penalty is unnecessary for its directly integrated trajectories. The default annual PINN seed is selected by training loss only. Learned parameters are frozen and inserted into the ODE solver, which integrates from 1900 through 1920. Neural network extrapolation and test-state restarts are not used.

1.4 Measured recovery and forecast results

Absolute relative parameter error is $100|\widehat{q}/q_{\text{true}} - 1|$. Table 1 reports mean and maximum errors across three seeds. The largest clean-case error was 0.0057%, compared with 0.796% for dense noisy data and 0.358% for annual noisy data. The selected annual estimate was $\widehat{r} = 1.003531$, $\widehat{a} = 1.997700$ and $\widehat{m} = 0.498663$.

The smaller annual error in this realization does not establish a sampling-frequency advantage. The finite noise realizations differ and the fixed constants are correct.

Table 1: Synthetic parameter recovery across optimizer seeds, conditional on correct fixed constants.

Training case	Rate	Mean error (%)	Max error (%)
dense clean	r	0.0032	0.0057
dense clean	a	0.0012	0.0021
dense clean	m	0.0014	0.0032
dense noisy	r	0.6402	0.6466
dense noisy	a	0.4856	0.4895
dense noisy	m	0.7895	0.7964
annual noisy	r	0.3453	0.3577
annual noisy	a	0.1197	0.1346
annual noisy	m	0.2736	0.2924

Table 2 reports five-point held-out RMSE and MAE separately for each species, against

both noisy observations and latent generating truth. Least squares has lower hare error against noisy observations, while the selected PINN has lower lynx error. The selected PINN is closer to latent truth for both species in this realization. This mixed result does not establish general superiority.

Table 2: Frozen ODE forecasts for five synthetic held-out years. H and P denote hare and lynx indices.

Method	Target	H RMSE	P RMSE	H MAE	P MAE
PINN ODE	Noisy	0.3319	0.3862	0.2780	0.2463
PINN ODE	Latent	0.0524	0.0820	0.0475	0.0713
Least squares	Noisy	0.2962	0.4412	0.2510	0.3035
Least squares	Latent	0.2236	0.1570	0.1910	0.1486

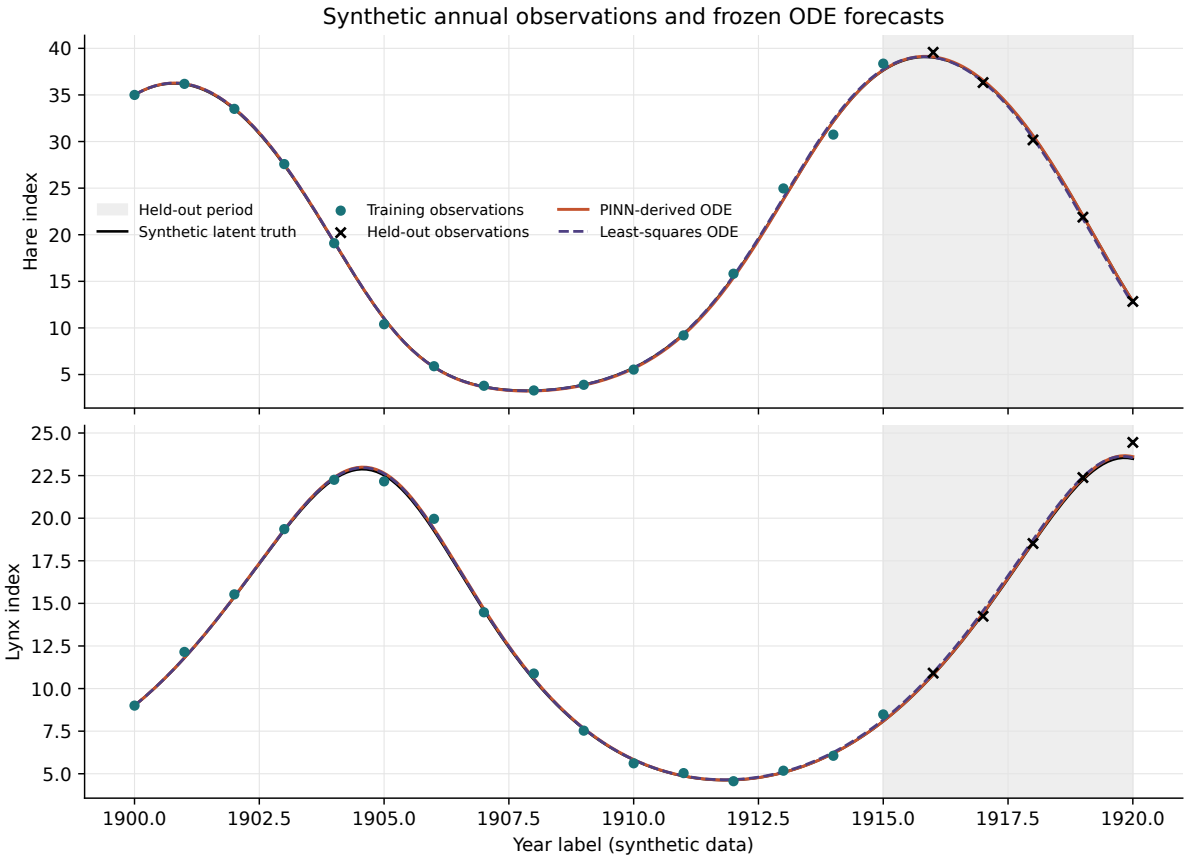


Figure 1: Synthetic observations and ODE forecasts. The shaded period is held out. Both methods integrate from the first training observation. The PINN seed is chosen using training loss only.

1.5 Sensitivity and interpretation

Prespecified annual-data runs vary physics weight from 10 to 1 or 100 and multiply initial rate guesses by 0.6 or 1.4, holding seed 17 fixed. Their forecasts are shown in Figure 3; these variants do not select a configuration on test errors. A local finite-difference sensitivity matrix of normalized training trajectories with respect to log parameters gives condition numbers 3.74 for varying r, a, m and 60.24 for varying all six parameters locally. The larger local problem contains

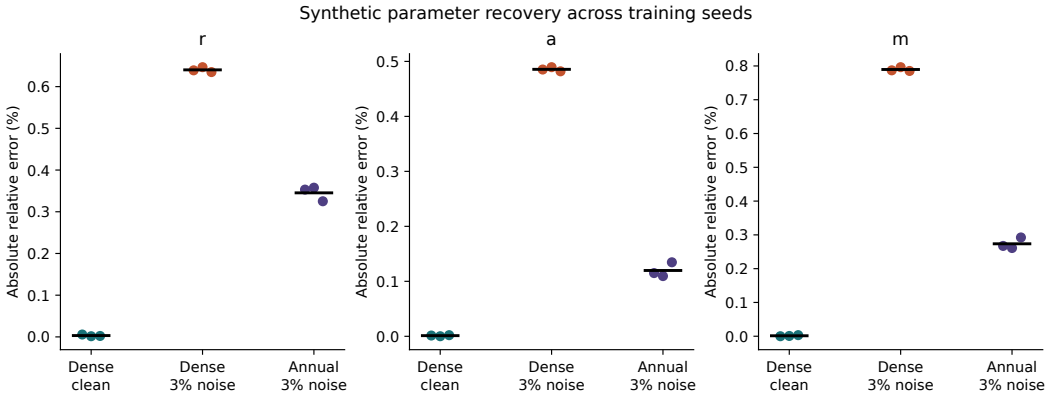


Figure 2: Absolute relative recovery errors. Points are optimizer seeds, and black marks are means. These are not confidence intervals.

less sensitive directions. This diagnostic is not an all-six PINN fit or a proof of structural or global identifiability.

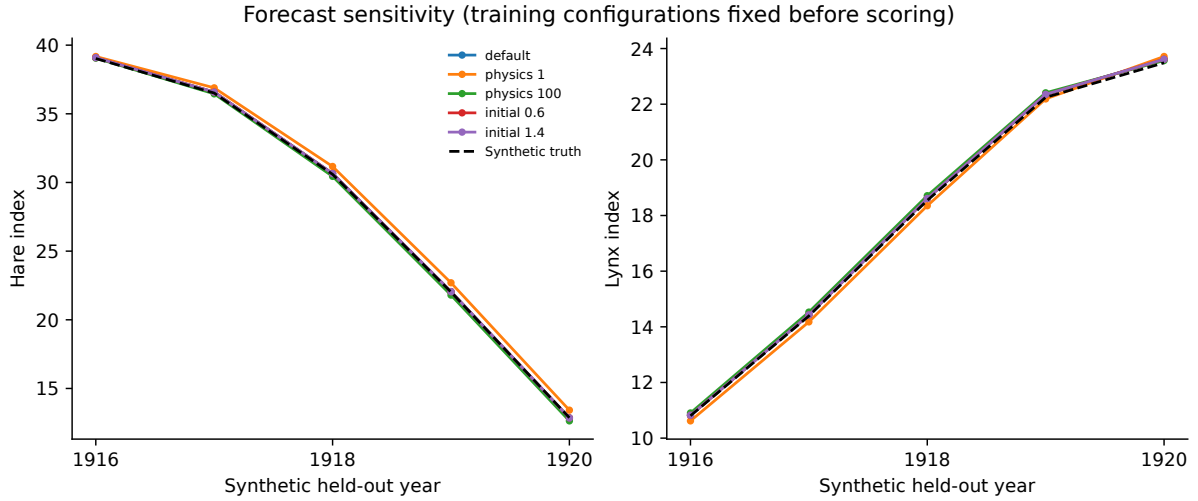


Figure 3: Sensitivity of synthetic forecasts to physics weight and initial guesses. Configurations were declared before held-out scoring.

The results rely on a correctly specified generating ODE, three known constants and an exact initial state. Finite collocation and nonconvex training may still leave a network/ODE mismatch, so residual checks and training-domain consistency plots are provided with the code. Three seeds on one noise realization do not quantify measurement uncertainty. Five held-out years and an idealized two-species system cannot address pelt trading bias, missing ecology or unknown observation scales. Historical fitting must use the team's agreed equations, units, feasible ranges and sealed test period.

1.6 Reproducible handoff

The repository supplies source code, configuration, an executed notebook, recovery tables, all default-seed forecast scores, EPS/PNG figures, optimization logs and a runtime/source-hash manifest. A three-column Year/Hare/Lynx CSV contains only the five held-out synthetic predictions for Advay. The solver and least-squares implementation are provisional scaffolding pending integration with Rishi and Advay. Yash's personal AI assistance log separates assistant-run checks from checks Yash still needs to perform.

A Supporting diagnostics

The following figures complete the numerical checks described in the main section. All inputs and results remain synthetic. Every plot has an included EPS source and a vector PDF companion for direct compilation with a PDF engine.

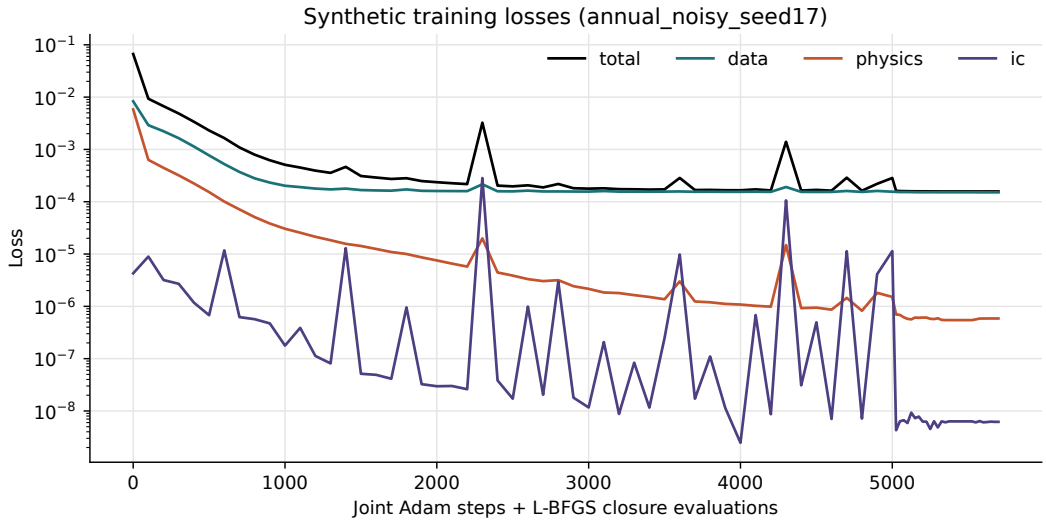


Figure 4: Weighted total loss and unweighted data, physics and initial-condition losses for the selected annual run. The horizontal coordinate combines joint Adam steps and subsequent L-BFGS closure evaluations. The warmup is omitted, and line-search trials need not be monotonic.

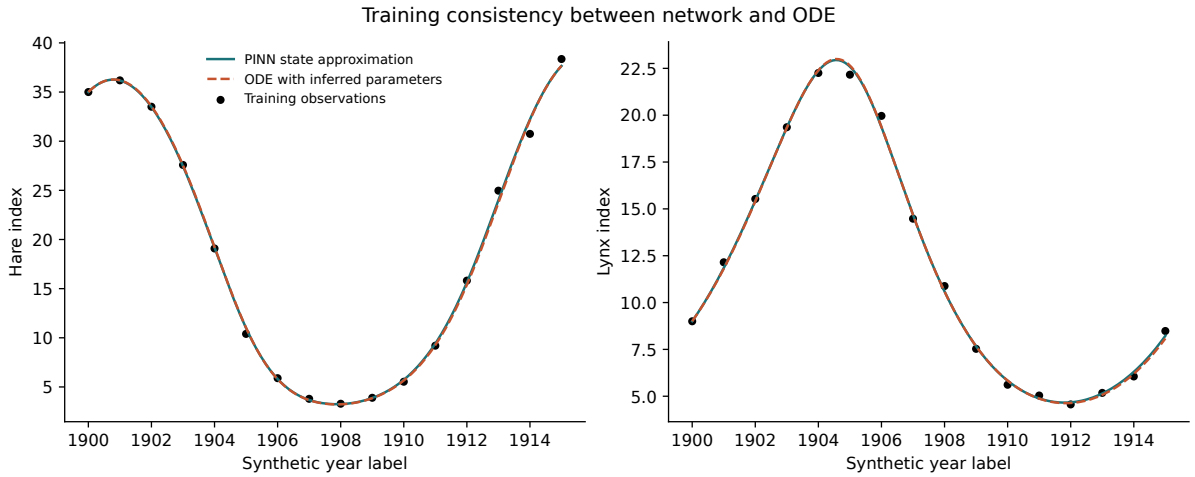


Figure 5: Neural predictions and the ODE trajectory with inferred parameters on the training interval. Their agreement checks whether fitting the network also produces a consistent mechanistic trajectory.

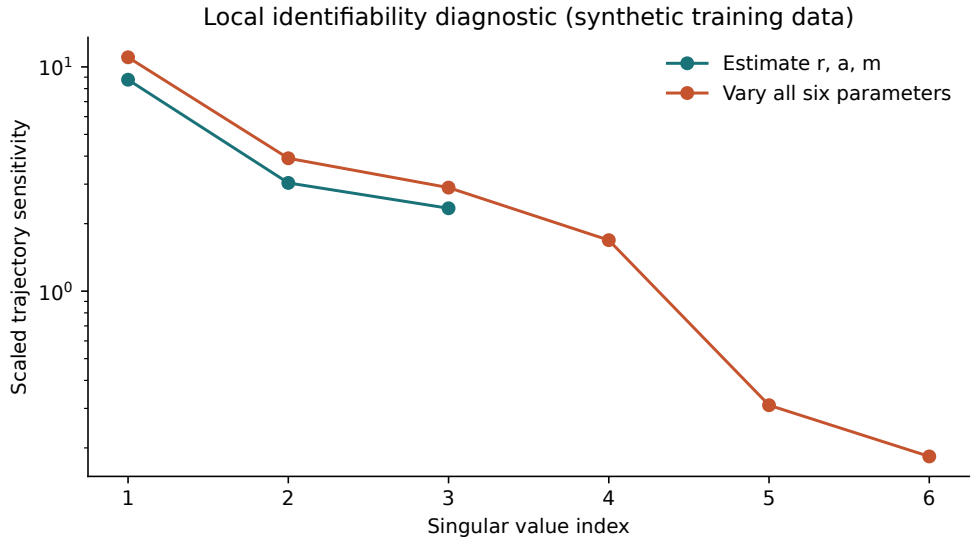


Figure 6: Singular values of the normalized training-trajectory sensitivity matrix with respect to log parameters, evaluated at the selected annual estimate. Locally varying all six parameters introduces less sensitive directions than varying only r, a, m. This is not an all-six PINN fit or a proof of global identifiability.

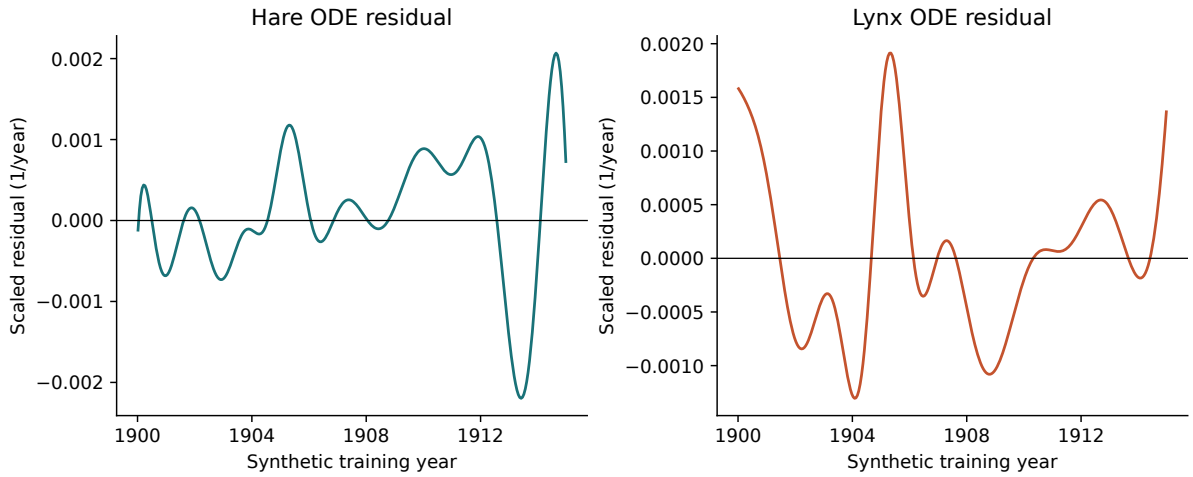


Figure 7: Hare and lynx residuals evaluated at 307 offset points in the training interval after fitting. These points diagnose error between collocation points and do not tune the model. Residual units are inverse years.

References

[1] PyTorch Contributors. torch.autograd.grad documentation. `https://docs.pytorch.org/docs/stable/generated/torch.autograd.grad.html`. Accessed 26 October 2026.

[2] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[3] Michael L. Rosenzweig and Robert H. MacArthur. Graphical representation and stability conditions of predator–prey interactions. *The American Naturalist*, 97(895):209–223, 1963.

[4] SciPy Contributors. scipy.integrate.solve_ivp documentation. `https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html`. Accessed 26 October 2026.